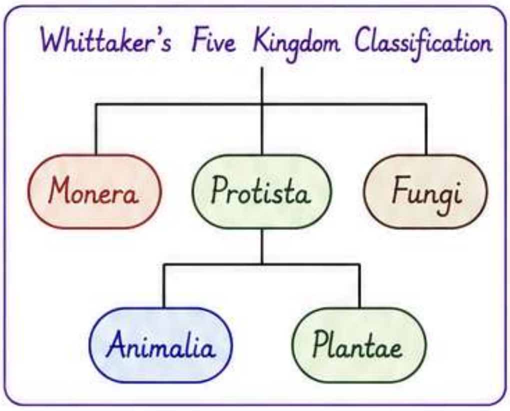
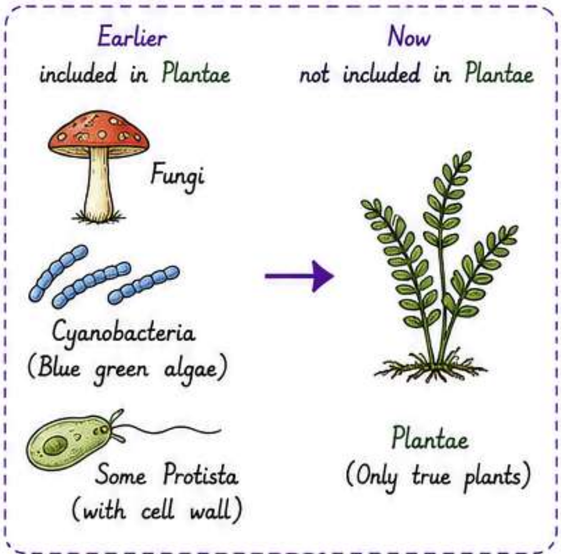
CHAPTER 3 : PLANT KINGDOM

INTRODUCTION

In the previous chapter, we looked at the broad classification of living organisms under the system proposed by Whittaker (1969) wherein he suggested the Five Kingdom classification viz. Monera, Protista, Fungi, Animalia and Plantae. In this chapter, we will deal in detail with further classification within Kingdom Plantae popularly known as the 'plant kingdom'.

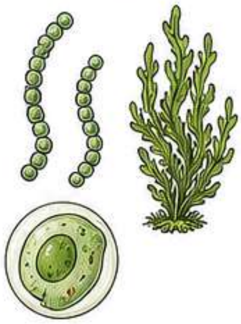


We must stress here that our understanding of the plant kingdom has changed over time.
Fungi and members of the Monera and Protista having cell walls have now been excluded from Plantae though earlier classifications put them in the same kingdom.
So, the cyanobacteria that are also referred to as blue green algae are not 'algae' any more.



In this chapter, we will describe Plantae under –

1. Algae



2. Bryophytes

3. Pteridophytes

4. Gymnosperms

5. Angiosperms

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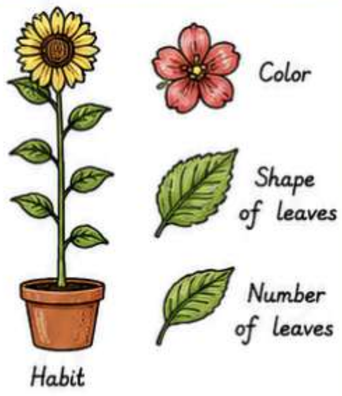
Classification within Angiosperms

1. Artificial system of classification

- Given by Linnaeus
- Based on superficial morphological characters such as habit, color, number and shape of leaves, etc.
- They were based on vegetative characters and androecium structures.
- Gave equal importance to vegetative and sexual characteristics.

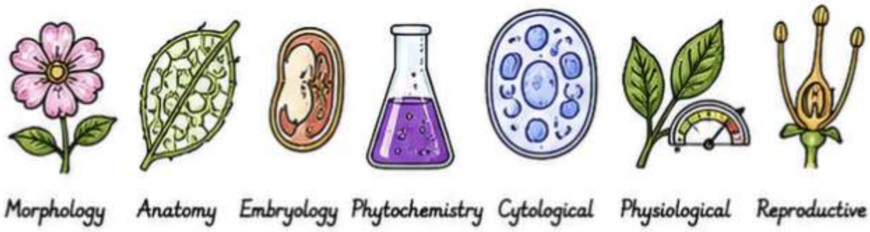


This is not acceptable since we know that often the vegetative characters are more easily affected by environment.

As against this Natural Classification System developed.

2. Natural system of classification

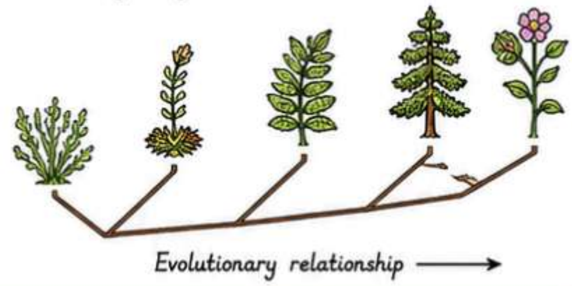
- Based on morphology, anatomy, embryology, phytochemistry, cytological, physiological and reproductive characters.
- Given by George Bentham and Joseph Dalton Hooker



3. Phylogenetic system of classification

– based on evolutionary relationship.

Given by Englar & Prantl



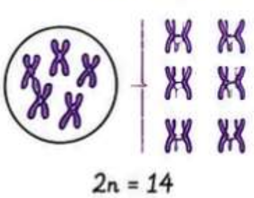
4. Numerical Taxonomy

- Based on all observable characteristics
- Numbers and codes assigned to all characters
- Easily carried out using computers

Character	Codes
Plant height	3
Leaf shape	2
Flower color	4
Fruit type	1
Seed number	5
....	......

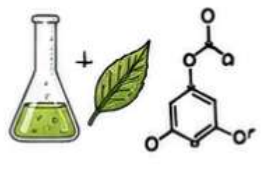
5. Cytotaxonomy

– Based on cytological information such as chromosome number, structure, behavior.



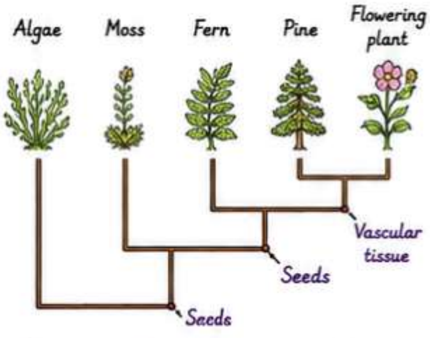
6. Chemotaxonomy

– Based on chemical constituents of plant to resolve doubts and confusions.



7. Cladistics taxonomy

– is an approach to biological classification in which organisms are categorized based on shared derived characteristics that can be traced to a groups most recent common ancestor and are not present in more distant ancestors.



(Shared derived characteristics)

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I) Division :- Algae

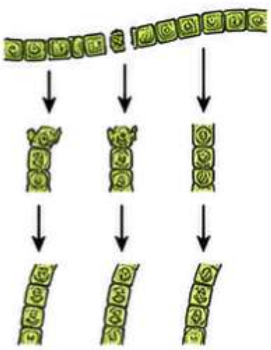
- Includes chlorophyll-bearing, simple, thalloid, autotrophic, and largely aquatic (freshwater and marine) organisms.
- They occur in a variety of other habitats such as moist stones, soils, wood.
- Some occur in association with fungi (lichens) and animals (on sloth bear).
- Size ranges from microscopic unicellular forms such as Chlamydomonas to colonial forms such as Volvox and to filamentous forms such as Ulothrix and Spirogyra.
- Massive plant-like bodies are seen in some marine forms (such as kelps).
- Algae reproduce by vegetative, asexual and sexual methods.

Habitats

Range of size and forms

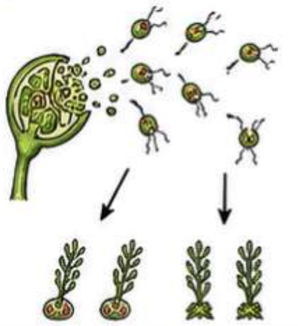
1. Vegetative reproduction

It is by fragmentation. Each fragment develops into a Thallus. (as in Spirogyra).



2. Asexual reproduction

It is by the production of different spores, called as zoospores. They are flagellated (motile) and on germination gives rise to new plants.

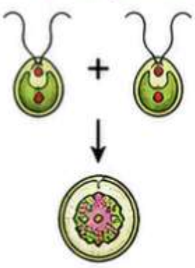


3. Sexual reproduction

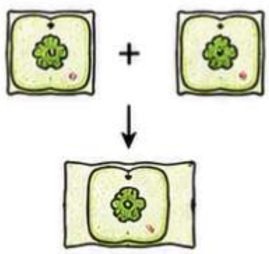
It takes place through fusion of two gametes.

- These gametes can be flagellated and similar in size (as in Chlamydomonas) or non-flagellated (non-motile) but similar in size (as in Spirogyra). Such reproduction is called isogamous.

(a) Isogamous (Chlamydomonas)

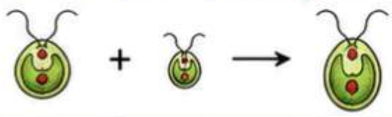


(b) Isogamous (Spirogyra)



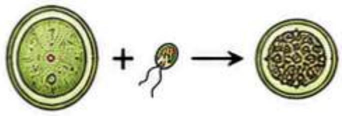
(c) Anisogamous

Fusion of two gametes dissimilar in size, as in some species of Chlamydomonas.



(d) Oogamous

Fusion between one large, non-motile (static) female gamete and a smaller, motile male gamete is termed oogamous, e.g., Volvox, Fucus.



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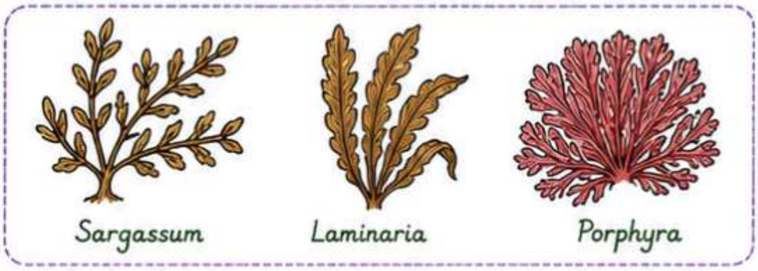
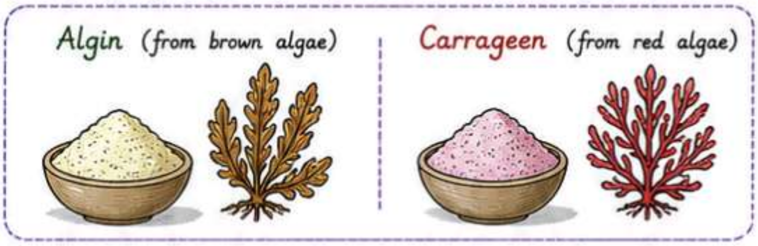
Economic Importance of Algae

- Carbon dioxide fixation on earth is majorly carried out by algae.

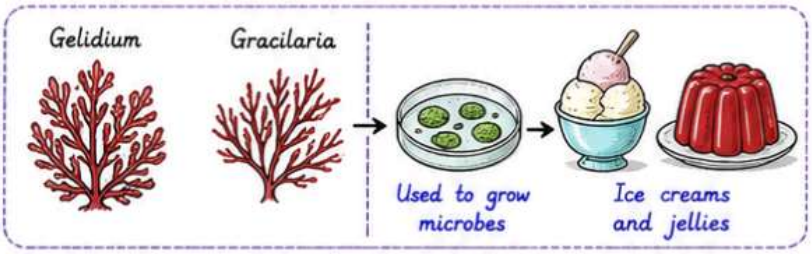
- Important as primary producers of energy-rich compounds.
- Many species of Sargassum, Laminaria, and Porphyra are marine algae used as food.



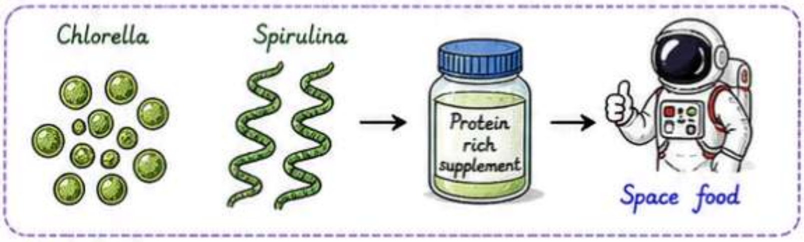
- Some brown and red algae species produce water-holding hydrocolloids. Example - Algin (brown algae) and carrageen (red algae).



- Agar obtained from Gelidium and Gracilaria is used to grow microbes and in preparation of ice creams and jellies.



- Chlorella and Spirulina are protein-rich unicellular algae, used as food supplements. They are also known as space food.



- The algae are divided into three main classes:-

i) Chlorophyceae
(Green algae)

Examples:
Chlamydomonas, Volvox, Spirogyra, Ulva

ii) Phaeophyceae
(Brown algae)

Examples:
Fucus, Sargassum, Laminaria

iii) Rhodophyceae
(Red algae)

Examples:
Porphyra, Gelidium, Gracilaria

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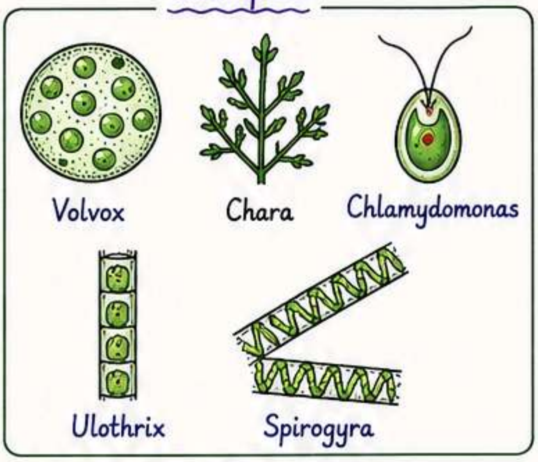
i) Chlorophyceae (Green algae)

- The members of Chlorophyceae are commonly called green algae.
- Occur in fresh water, brackish water and salt water.
- The plant body may be unicellular, colonial, or filamentous.
- They are usually Grass green in color due to abundance of pigment chlorophyll a and b.
- The chloroplasts may be discoid, plate like, reticulate, cup shaped, spiral or ribbon shaped in different sps.
- Chloroplast of most of the Chlorophyceae contains pyrenoids.

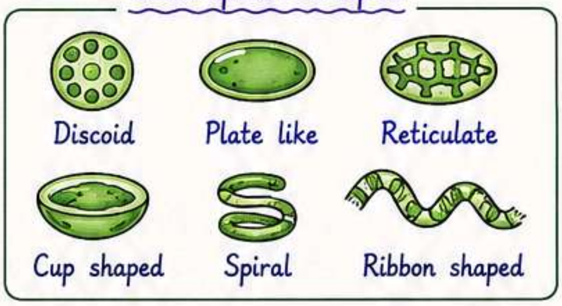
> Pyrenoids – Are storage bodies containing proteins in addition to starch.

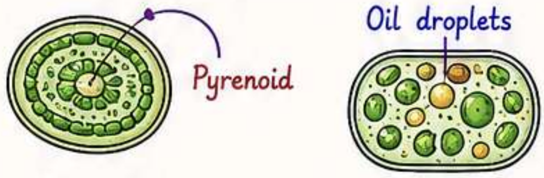
- Food storage occurs in the form of oil droplets in some algae.
- Cells have rigid cell wall: inner layer made of cellulose, outer layer made of pectose.
- Examples include Volvox, Chara, Chlamydomonas, Ulothrix and Spirogyra.
- Vegetative reproduction takes place by fragmentation.
- Asexual reproduction takes place by flagellated (2–8) Zoospores produced in Zoosporangia.
- The sexual reproduction shows variation in the type & formation of sex cells and may be Isogamous, Anisogamous, or Oogamous.

Examples

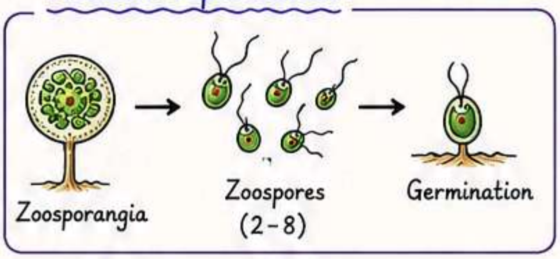


Chloroplast shapes

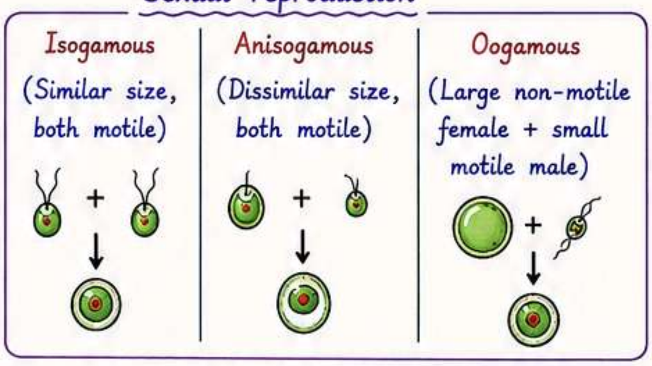




Asexual reproduction



Sexual reproduction



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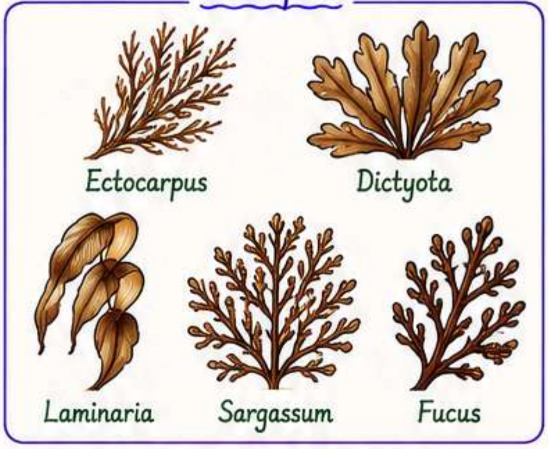
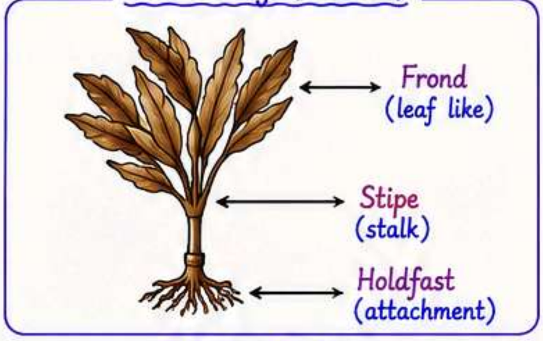
ii) Phaeophyceae (Brown algae)

- The member of Phaeophyceae or brown algae is primarily marine forms.
- They show great variation in size and form.
- They range from simple-branched, filamentous forms (Ectocarpus) to profusely branched forms such as kelps (may reach a height of 100 m).
- Possess chlorophyll a, c, c, carotenoids, and xanthophylls.
- Vary in colour from olive green to various shades of brown (depending on amount of xanthophylls and fucoxanthin).
- Food stored as complex carbohydrates such as laminarin or mannitol.
- Vegetative cells have cellulosic wall covered on the outside by gelatinous coating of algin.
- Cell contains a centrally located vacuole and nucleus in addition to plastids.
- The plant body is usually attached to the substratum by a holdfast and has a stalk the Stipe and leaf like photosynthetic organ Frond.
- Vegetative reproduction takes place by fragmentation.
- Asexual reproduction in most brown algae is by biflagellate zoospores that are pear-shaped and have two unequal laterally attached flagella.
- Sexual reproduction may be isogamous, anisogamous or oogamous.
- Union of gametes takes place in water or within oogonium (oogamous species).
- The gametes are pyriform (pear-shaped) and bear two laterally attached flagella.
- Example – Ectocarpus, Dictyota, Laminaria, Sargassum, and Fucus.

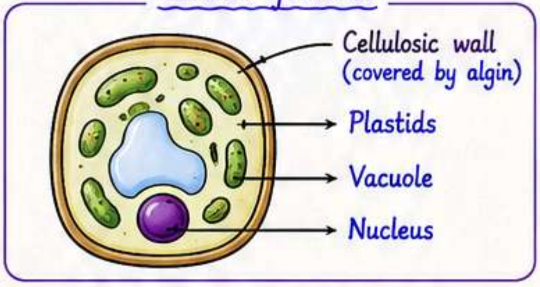
Examples



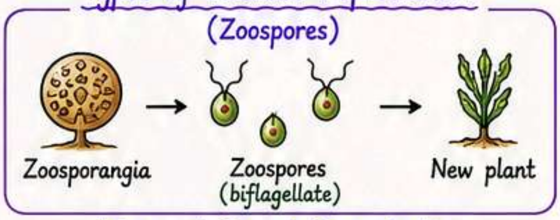
Plant body (Thallus)



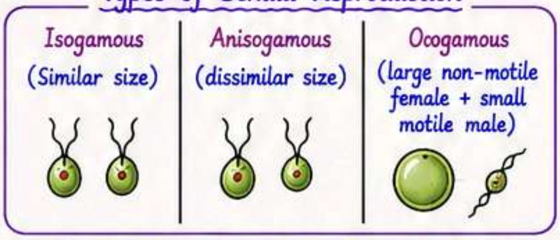
Cellular features



Types of Asexual Reproduction



Types of Sexual Reproduction



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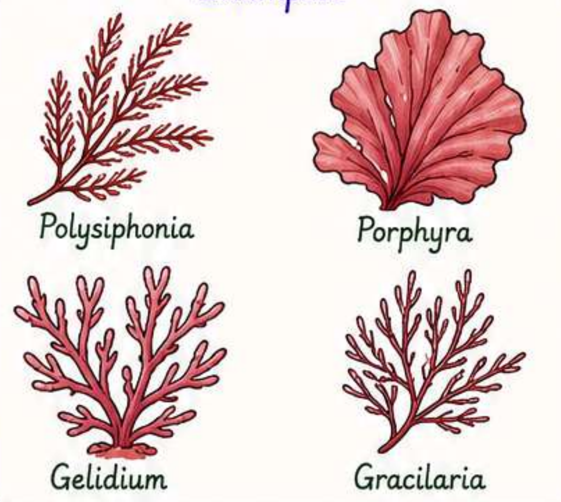
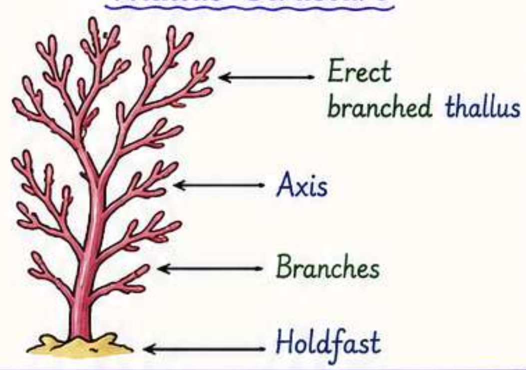
ii) Rhodophyceae (Red algae)

- Commonly called red algae due to the presence of red pigment, r-phycoerythrin. Also contains chlorophyll a and d.
- Mainly marine forms with bulk mass inhabiting warmer areas.
- Occur in well-lighted regions i.e., close to the surface of water and also in deeper areas.
- Red thalli of most of these species are multicellular. Some have complex body organization.
- Cell wall is made up of cellulose.
- Food is stored as Floridian starch similar to amylopectin and glycogen in structure.
- The red algae usually reproduce vegetatively by fragmentation.
- They reproduce asexually by non-motile spores and sexually by non-motile gametes.
- Sexual reproduction is oogamous. Complex post fertilization developments.
- Example – Polysiphonia, Gelidium, Gracilaria, Porphyra.

Examples



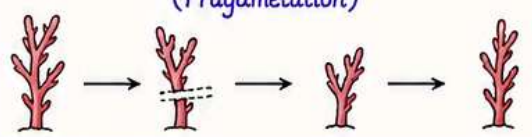
Thallus Structure



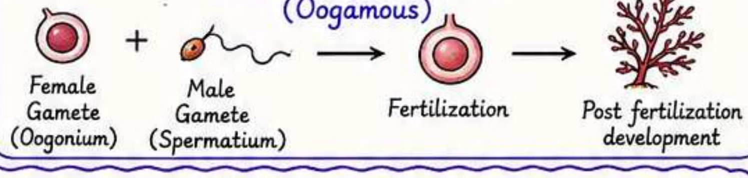
Storage Food

Floridian starch
(stored in cytoplasm)

Sexual Reproduction
(Fragametation)



Sexual Reproduction
(Oogamous)



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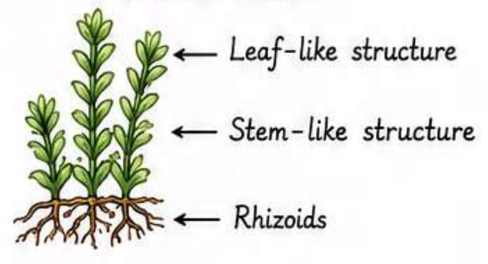
II) Division: – Bryophyta

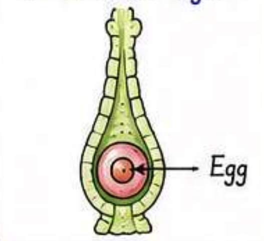
- Bryophytes are also called amphibians of plant kingdom since they live on land, but depend on water for sexual reproduction.
- Usually occur in cool, damp, and shady areas.
- Play an important role in plant succession on bare rocks/soils.
- Plant body more differentiated than algae.
- Plant body is thallus-like and prostrate or erect and attached to substratum by unicellular or multicellular rhizoids.
- They lack true roots, stem and leaves; may possess root-like, stem-like, and leaf-like structures.
- The plant body of the Bryophytes is haploid. It produces gametes, hence is called Gametophyte.
- The sex organs in bryophytes are multicellular.
- The male sex organ is called Antheridium. It produces biflagellate Antherozoid.
- The female sex organ is called Archegonium. It is flask shaped and produces a single egg.
- The antherozoids are released into the water where they come in contact with Archegonium. An antherozoid fuses with egg to produce Zygote. (Zygote do not undergo reduction division i.e. meiosis immediately.)
- The zygote grows into a multicellular body called Sporophyte.
- Sporophyte is dependent on gametophyte for food. Hence, it remains attached to the gametophyte.
- Few cells of sporophyte undergo meiosis to produce haploid spores.
- Spores germinate to form gametophyte.

Bryophyte Plant

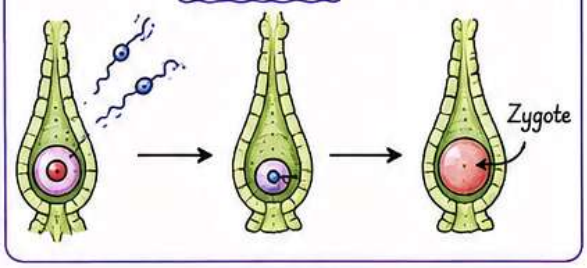


Antheridium (Male sex organ)

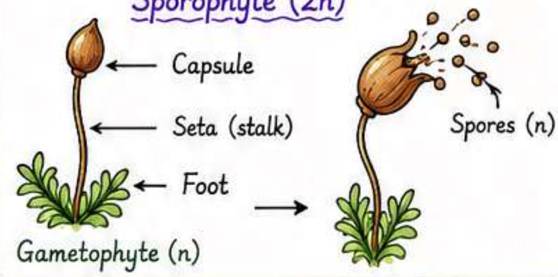
Archegonium (Female sex organ)



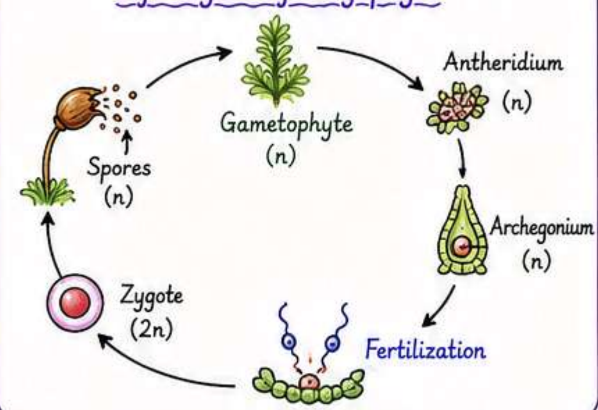
Fertilization



Sporophyte (2n)



Life Cycle of Bryophyte



Classes of bryophytes

Bryophytes are divided into two classes: –
1) liverworts 2) Mosses

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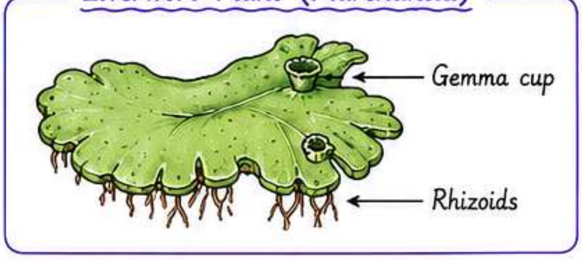
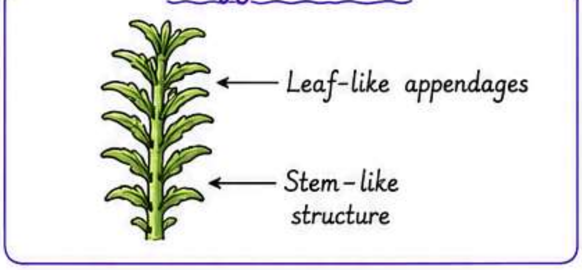
Class: - 1. Liverworts

- Grow in moist, shady habitats such as banks of streams, marshy ground, damp soil, bark of the tree and deep in the wood.
- Plant body of the liverwort is thalloid, e.g., Marchantia.
- The thallus is dorsiventrally appressed to the substrate.
- Leafy members have tiny leaf-like appendages in two rows on the stem-like structures.
- Asexual reproduction takes place by fragmentation of thalli, or by the formation of specialized structure called Gemmae (sing: - Gemma).
- The Gemmae are green, multicellular, asexual buds, which develops in small receptacles called Gemma cups located on the thalli.
- The gemmae become detached from the plant body and germinate to form new individual.
- In sexual reproduction, male and female sex organs are produced either on the same or different thalli.
- The sporophyte is differentiated into a foot, seta and capsule.
- After meiosis spores are produced within the capsule. These spores germinate to form free-living gametophyte.

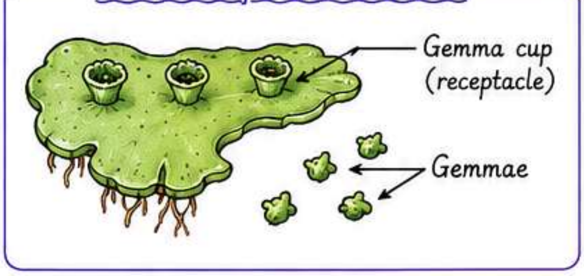
Liverwort Plant (Marchantia)



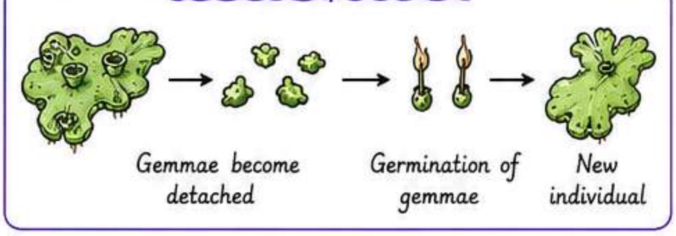
Leafy Liverwort



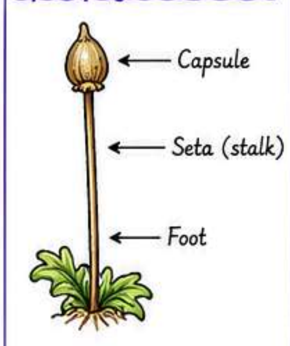
Gemma Cups and Gemmae



Asexual Reproduction

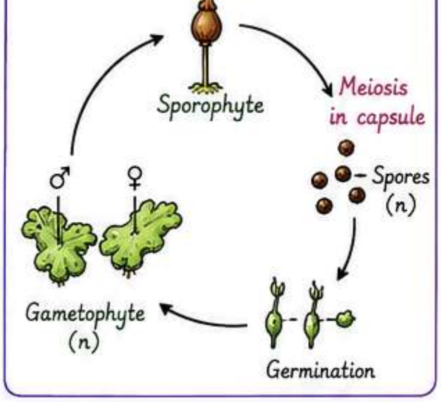


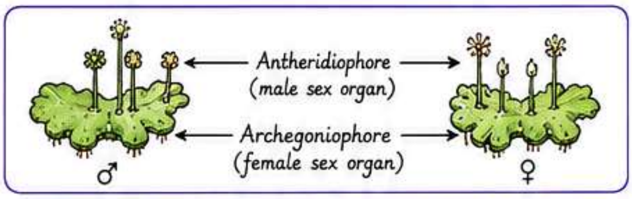
Sporophyte Structure



(Attached to gametophyte)

Life Cycle





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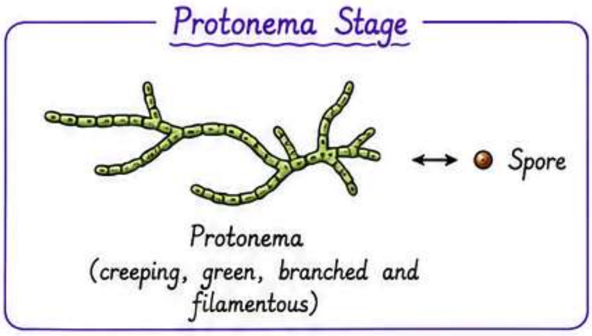
Class: - 2. Mosses

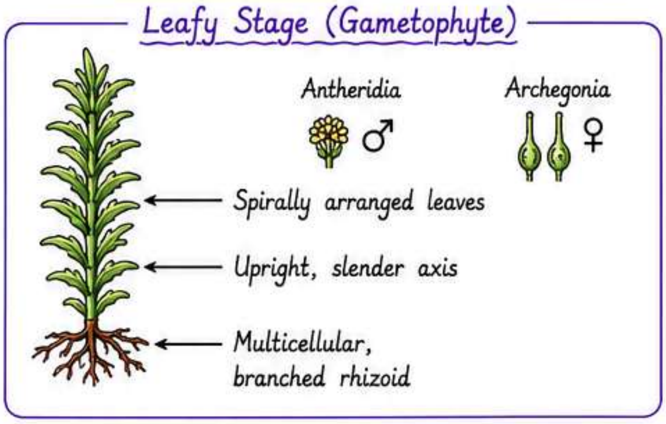
- The predominant stage of the life cycle of a moss is the gametophyte.

Gametophyte

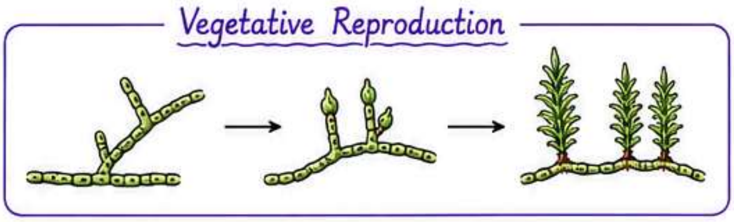
- Consist of two stages.
- The first stage is the protonema stage, which develops directly from the spore.
  - It is a creeping, green, branched and frequently filamentous stage.



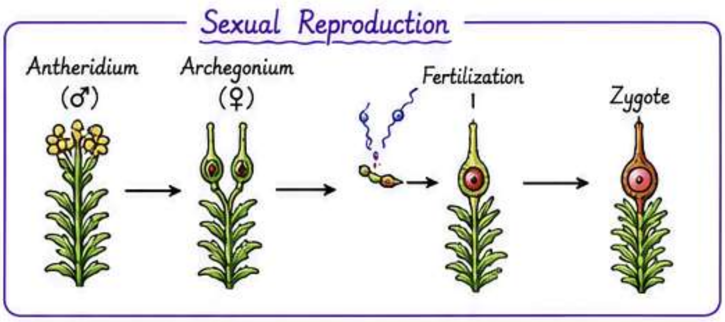
- The second stage is the Leafy stage, which develops from the secondary protonema as a lateral bud.
  - It consists of upright, slender axes bearing spirally arranged leaves.
  - It is attached to the soil through multicellular and branched rhizoid.
  - This stage bears sex organ.



- Vegetative reproduction in mosses is by fragmentation and budding in the secondary protonema.

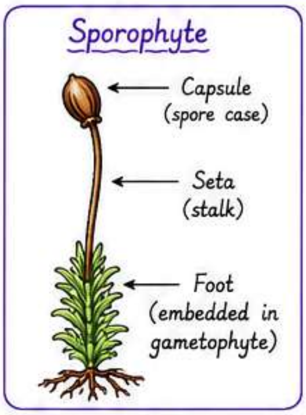


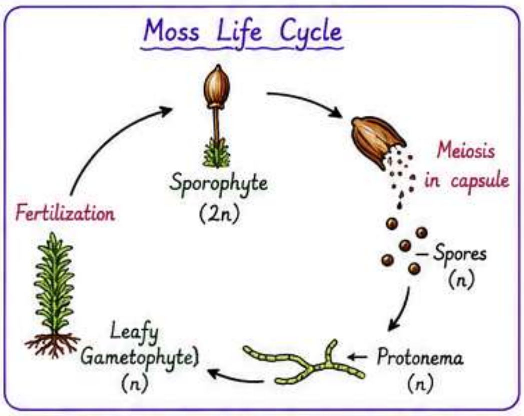
- In sexual reproduction, the sex organ antheridia and archegonia are produced at the apex of the leafy shoots.
- After fertilization, the zygote develops into a sporophyte.



Sporophyte

- More elaborate than liverworts.
- Consists of foot, seta, and capsule.
- Capsule contains spores.
- Spores formed by meiosis.
- Elaborate mechanism of spore dispersal.
- Example - Funaria, Polytrichum, and Sphagnum.





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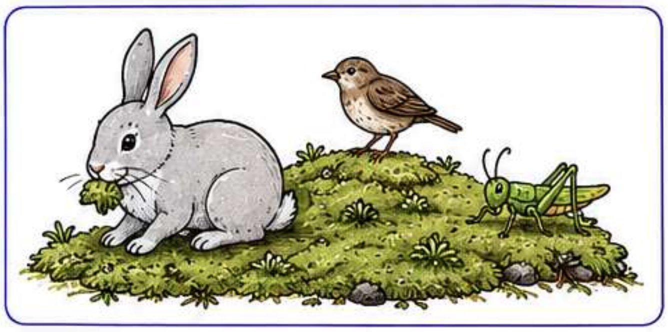
CHAPTER 3 : PLANT KINGDOM

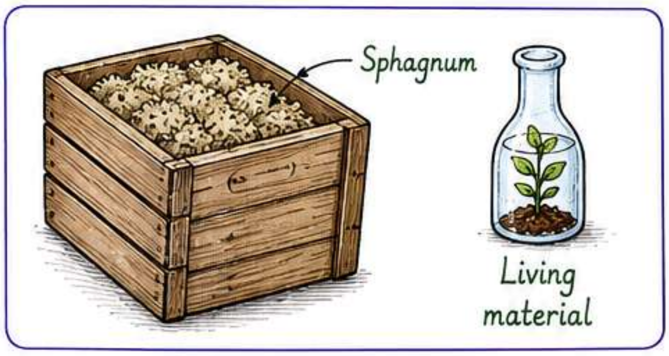
Economic Importance

- Provide food for herbaceous mammals, birds, and insects.

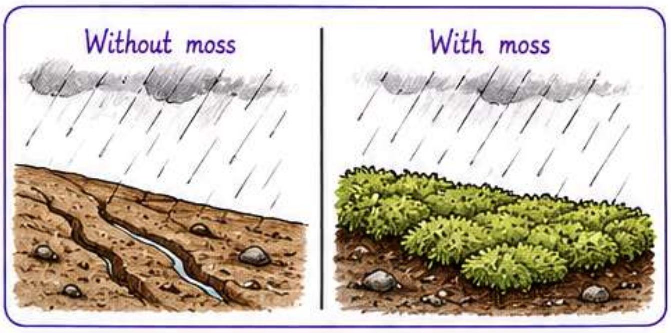


- Peat provided by Sphagnum (a moss) is used as fuel.

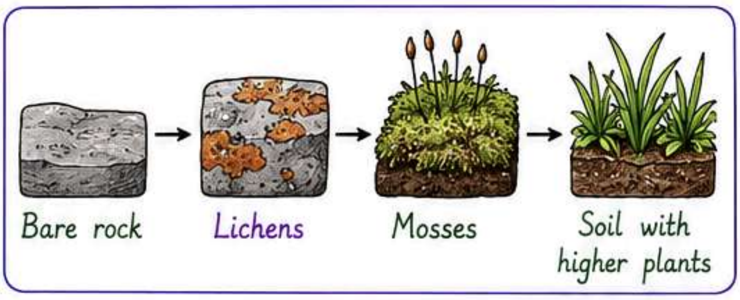
- Sphagnum is also used as packing material in trans-shipment of living material because of their water-holding capacity.



- They form dense mats on the soil and hence prevent soil erosion.



- Mosses along with lichens form the pioneer community in land and desert succession.



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CHAPTER 3 : PLANT KINGDOM

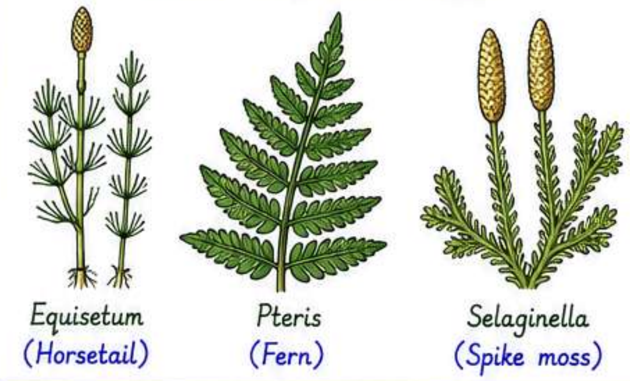
III) Division: - Pteridophyta

The Pteridophytes include Horsetails and Ferns. The Pteridophytes are used for medicinal purpose and as soil-binders. They are also frequently grown as ornamentals.

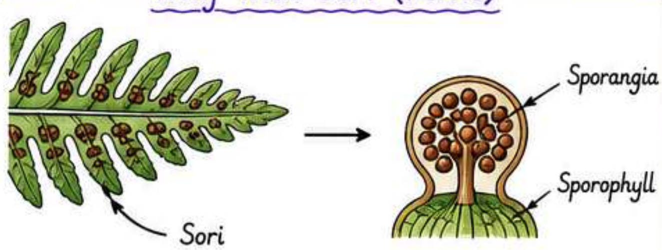
General Characteristics

- The Pteridophytes are found in cool, damp, shady places.
- Evolutionary the Pteridophytes are the first terrestrial plants to possess vascular tissues - xylem and phloem.
- In Pteridophytes the dominant phase in the life cycle is sporophytic plant body.
- They have well-differentiated true stem, leaves, and roots.
- Leaves may be small (microphylls) as in Selaginella or large (macrophylls) as in ferns.
- Sporophytes bear sporangia, which develop in association with leaves called sporophylls.
- In some Pteridophytes, sporophylls form distinct, compact structures called strobili or cones (Selaginella, lycopodium, Equisetum).
- Sporangia produce spores by meiosis in spore mother cells.
- Spores germinate to form small, multicellular, free-living photosynthetic thalloid gametophyte called prothallus.

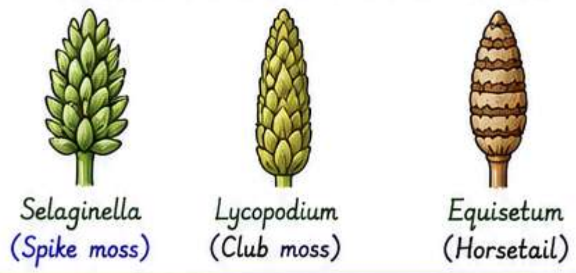
Examples of Pteridophytes



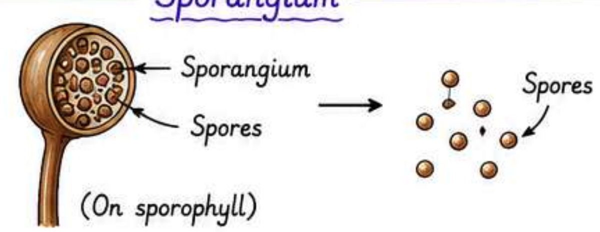
Leaf with Sori (Ferns)



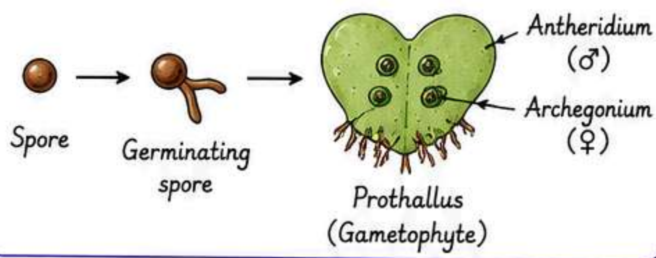
Strobili (Cones) in Pteridophytes



Sporangium



Germination of Spore



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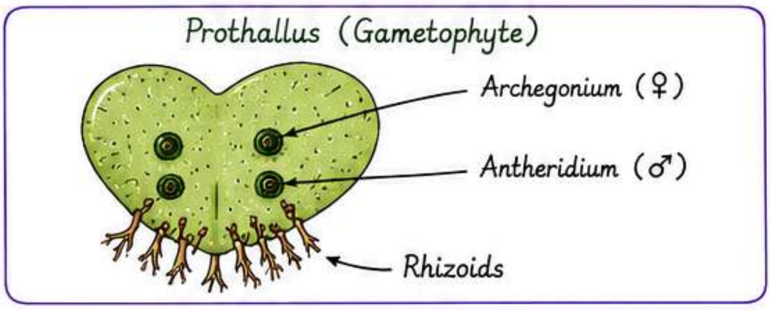
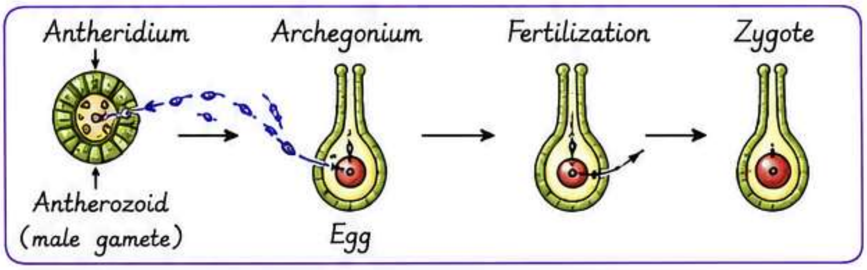
CHAPTER 3 : PLANT KINGDOM

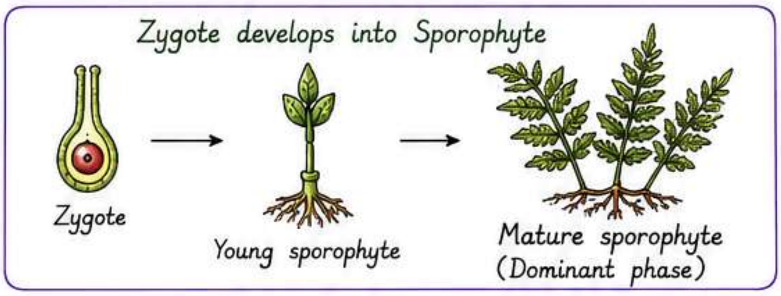
Prothallus (Gametophyte) & Sporophyte

Prothallus (Gametophyte)

- Require cool, damp, shady places to grow.
- Also require water for sexual reproduction.
- The gametophytes bears male and female sex organ called Antheridia and Archegonia, respectively.
- Fusion of male gamete (Antherozoid) with the egg present in the Archegonium results in the formation of Zygote.
- Zygote produces well-differentiated, multicellular sporophyte (dominant phase of the Pteridophytes.)

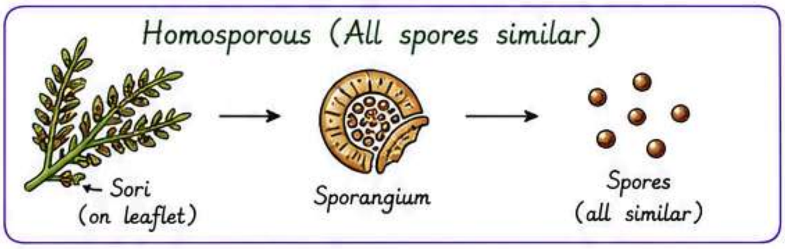


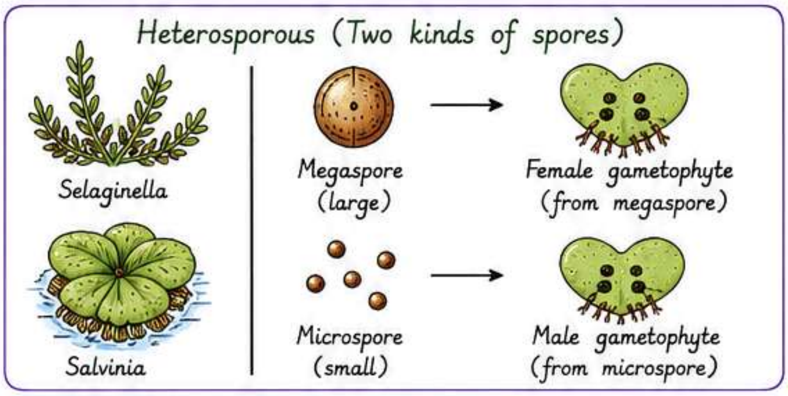


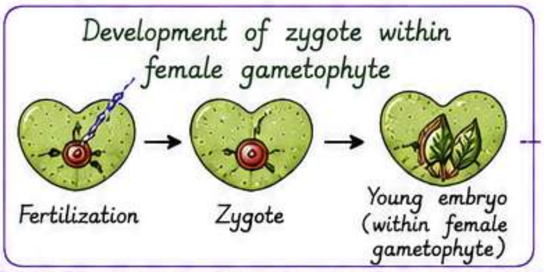


Sporophyte

- In majority of the Pteridophytes all the spores are of similar kinds; such plants are called Homosporous.
- Genera like Selaginella and Salvinia which produce two kinds of spores, large (Macrospore) and small (Microspore) are called Heterosporous.
- The megaspores and microspores germinate and give rise to female and a male gametophyte, respectively.
- The female gametophytes in these plants are retained on the parent sporophyte for variable periods.
- The development of the zygotes into the young embryos takes place within the female gametophytes.
  This step is a precursor to the seed habit considered an important step in evolution.





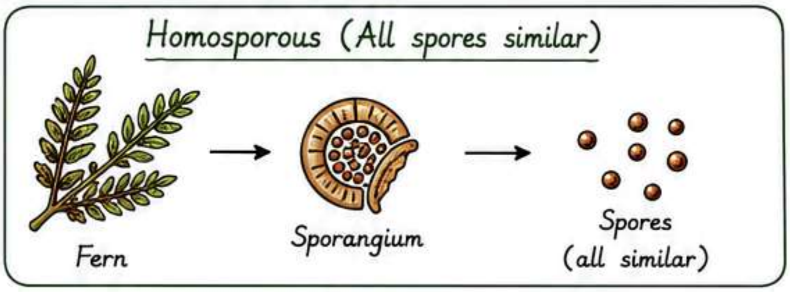


This step is a precursor to the seed habit considered an important step in evolution.

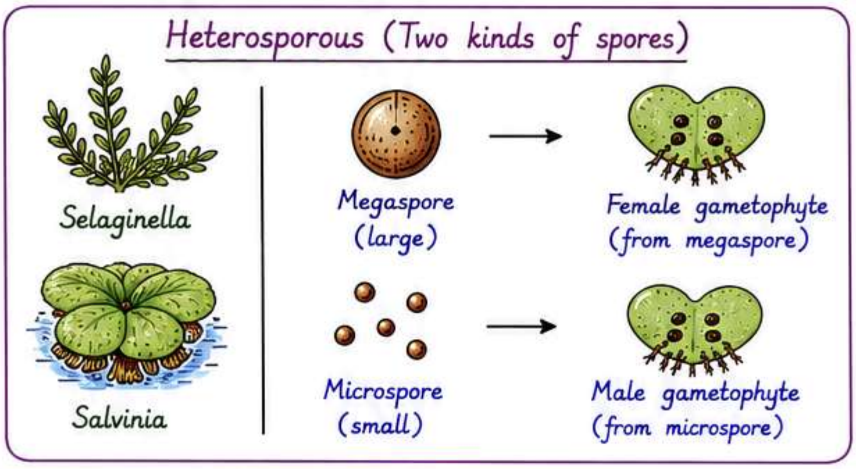
CHAPTER 3 : PLANT KINGDOM

Sporophyte

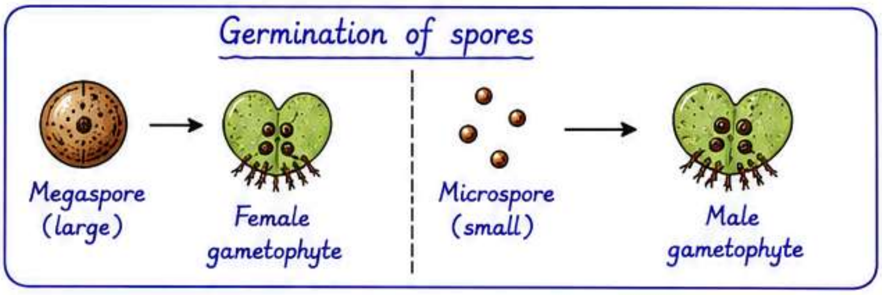
① In majority of the Pteridophytes all the spores are of similar kinds; such plants are called Homosporous.



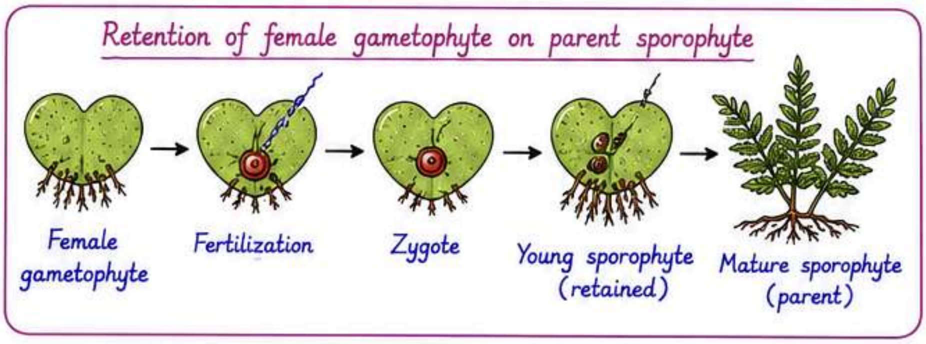
② Genera like Selaginella and Salvinia which produce two kinds of spores, large (Macrospore) and small (Microspore) are called Hetresporous.



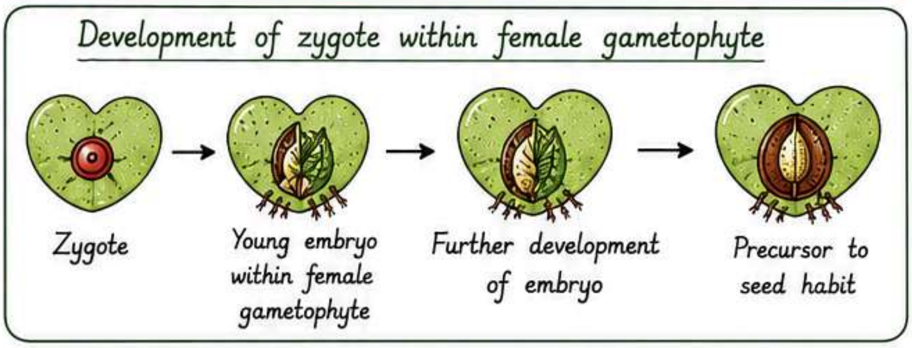
③ The megaspores and microspores germinate and give rise to female and a male gametophyte, respectively.



④ The female gametophytes in these plants are retained on the parent sporophyte for variable periods.



⑤ The development of the zygotes into the young embryos takes place within the female gametophytes. This step is a precursor to the seed habit considered an important step in evolution.



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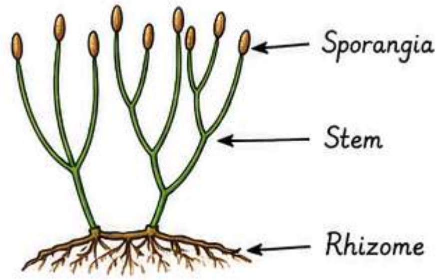
CHAPTER 3 : PLANT KINGDOM

Classification of Pteridophytes

The Pteridophytes are classified into four classes.

1. Psilopsida (Whisk Fern)

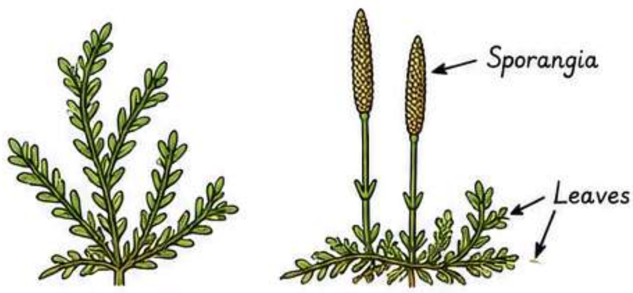
- These are the simplest Pteridophytes.
- They have naked stems with whorled, dichotomously branched leaves.
- Example – Psilotum



Psilotum

2. Lycopsida (Club Mosses)

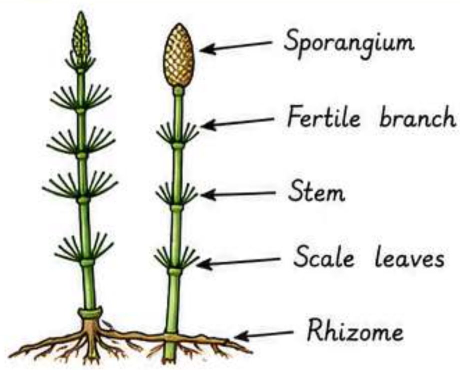
- They are small herbaceous plants.
- Leaves are small and scale-like, arranged spirally or in four rows.
- They have microphylls with a single vascular strand.
- Example – Selaginella, Lycopodium



Selaginella Lycopodium

3. Sphenopsida (Horsetail)

- They are the horsetails.
- Stems are jointed and have silica deposits.
- Leaves are reduced to small scales arranged in whorls.
- Sporangia form at the tips of fertile branches.
- Example – Equisetum



Equisetum

4. Pteropsida (Ferns)

- These are true ferns.
- They have large, divided leaves called fronds.
- Sporangia are borne on the underside of leaves in sori.
- Example – Dryopteris, Pteris, Adiantum, Salvinia, Marsilea

Dryopteris Pteris Adiantum Salvinia Marsilea

CHAPTER 3 : PLANT KINGDOM

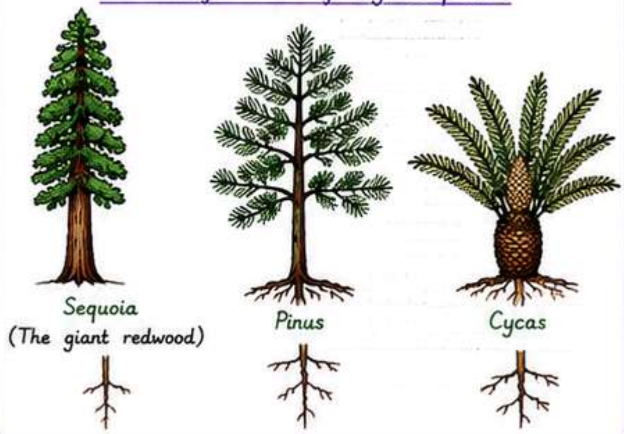
IV) Division : – Gymnospermae

- Word gymnosperms, gymnos – naked, sperma – seeds
- Ovules not enclosed by any ovary wall
- Seeds formed after fertilization are not covered (i.e., naked).
- Include medium-sized trees, shrubs, and tall trees
- Contains the world's largest plant Sequoia – the giant redwood
- Plants have tap roots. Roots in some genera show symbiotic associations.

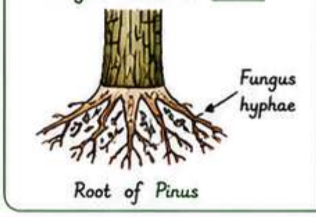
> 1. Mycorrhiza shows association of fungi with Pinus roots.
> 2. Coralloid roots of Cycas show association with N_2-fixing Cyanobacteria.

- The stems could be branched (Pinus, Cedrus) or unbranched (Cycas)
- The leaves may be simple (Pinus) or compound (Cycas)
- The leaves in gymnosperms are well-adapted to withstand extreme temperature, humidity and wind. In conifers, needle-like leaves with thick cuticle and sunken stomata reduce surface area and help to reduce water loss.

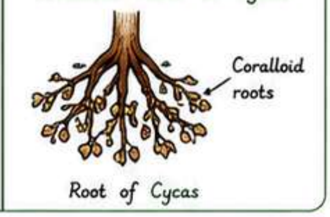
General features of Gymnosperms



Mycorrhiza in Pinus

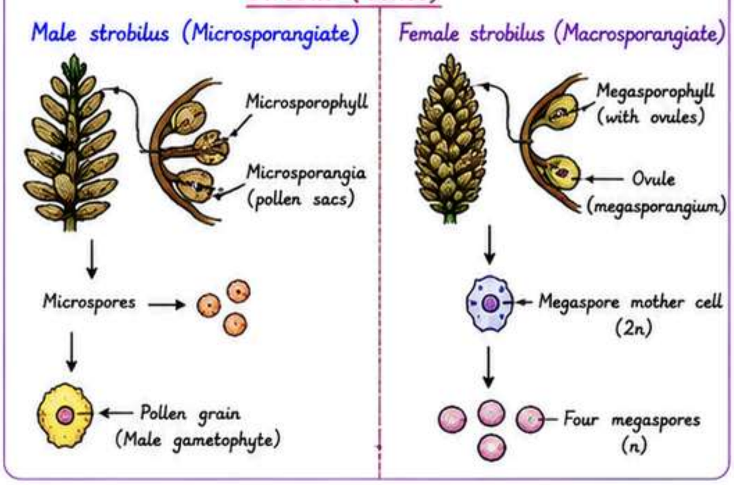


Coralloid roots in Cycas



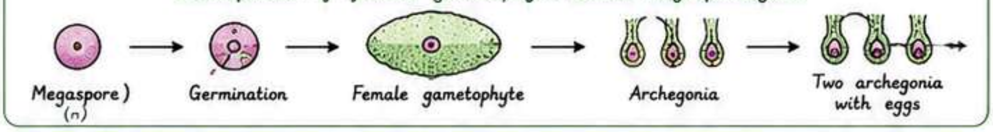
1. Gymnosperms are Hetresporous.
2. They produce haploid microspores and megaspore within sporangia that are borne on sporophylls which are arranged spirally along the axis to form lax or compact Strobilli or cone.
3. The strobili bearing only microsporophylls and microsporagia are called Microsporangiate or male strobili.
4. The microspores in the microsporangia develop into a male gametophyte generation which is reduced and is confined to only a limited number of cells. This reduced gametophyte is called a pollen grain and it develops within the microsporangia.

Strobili (Cones)



5. The strobili or cones bearing megasporophylls with ovules or megasporangia are called Macrosporangiate or female strobili.
6. Male and female strobili may be borne on same tree (Pinus) or on different trees (Cycas).
7. Megaspore mother cell divides mieotically to form four megaspores.
8. Megaspore mother cell is a differentiated cell of nucellus. Nucellus protected by envelopes is known as an ovule.
9. One of the megaspore enclosed within the megasporangium (nucellus) develops into a multicellular female gametophyte that bears two or more female sex organ or Archegonia which remains within megasporangium.
10. Male and female gametophytes do not have independent existence, hence remain within sporangia.

Development of female gametophyte within megasporangium



> **Gymnosperms**
> Male and female gametophytes do not have independent existence, hence remain within sporangia.

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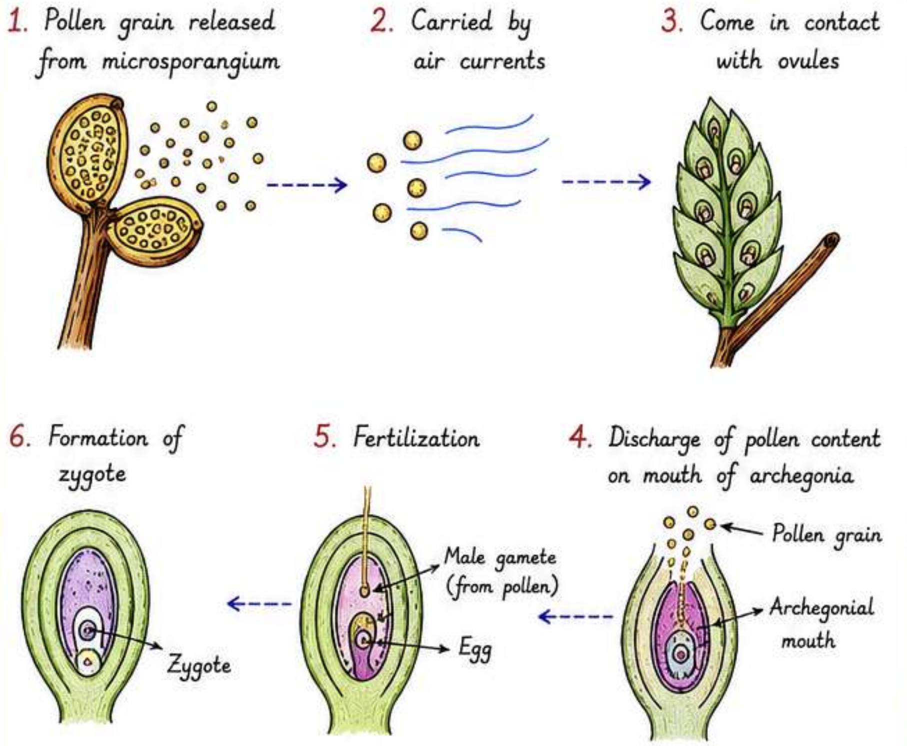
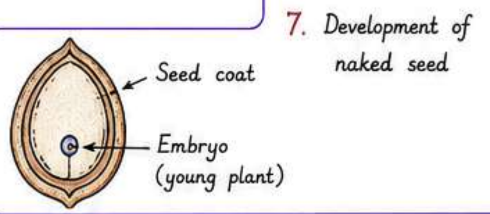
CHAPTER 3 : PLANT KINGDOM

IV) Division : – Gymnospermae

Steps in fertilization :

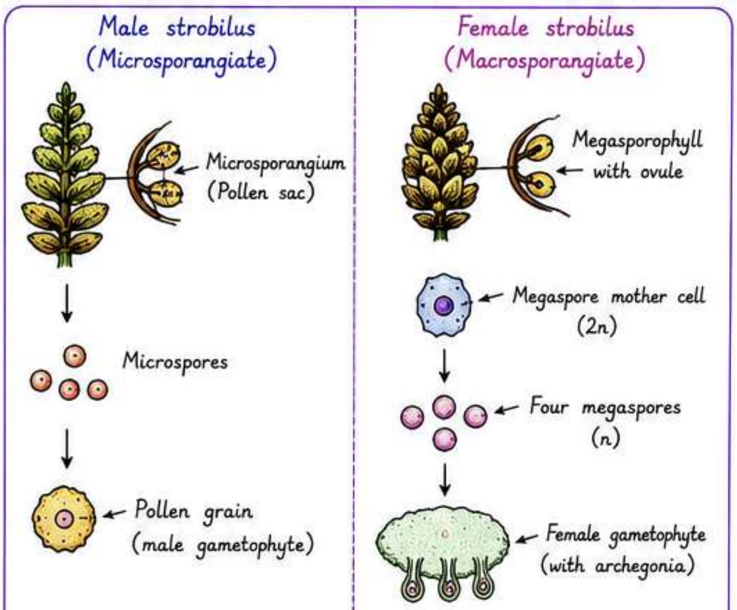
Pollen grain released from microsporangium
↓
Carried by air currents
↓
Come in contact with ovules
↓
Discharge of pollen content on mouth of archegonia
↓
Fertilization
↓
Formation of zygote
↓
Development of naked seed





Summary of key points

- ★ Gymnosperms are Hetresporous.
- ★ Male strobili (Microsporangiate) produce microspores which develop into pollen grains (male gametophyte).
- ★ Female strobili (Macrosporangiate) bear ovules (megasporangia) in which megaspore develops into female gametophyte.
- ★ Fertilization occurs within the archegonium of the ovule.
- ★ Seeds are naked as they are not enclosed by an ovary wall.



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CHAPTER 3 : PLANT KINGDOM

Gymnosperms are classified into three classes.

1. Cycadopsida (cycadophyta).

Ex - Cycas

- Plants are tree like, unbranched.
- Leaves large, pinnately compound.
- Male and female strobili are large.
- Tap root with coralloid roots having symbiotic cyanobacteria.

Cycas

Male strobilus

Female strobilus

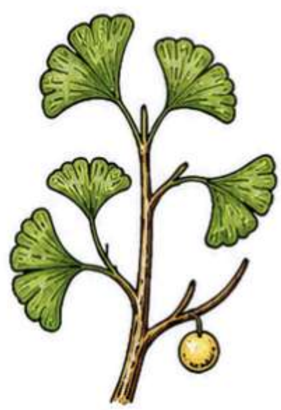
2. Coniferopsida (conifers).

Ex - Pinus, Sequoia, Ginkgo (Living fossil)

- Plants are usually tall trees.
- Leaves needle like or scale like.
- Wood is soft, resinous.
- Male and female cones are usually separate but borne on same tree (e.g., Pinus).

Pinus

Sequoia (Giant redwood)



Ginkgo (Living fossil)

3. Gnetopsida.

Ex - Gnetum.

- Mostly shrubs or climbers.
- Leaves opposite, broad with reticulate venation.
- Vessels are present in xylem.
- Male and female strobili are simple.

Gnetum

Male strobilus

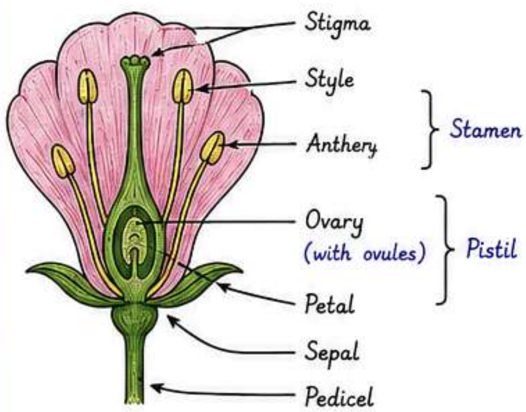
Female strobilus

CHAPTER 3 : PLANT KINGDOM

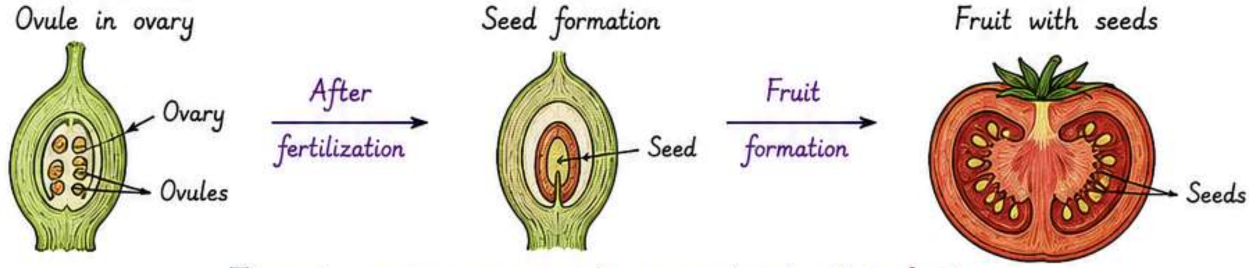
Division :- Angiosperms (Flowering plant)

- Unlike the gymnosperms where the ovules are naked, in the angiosperms or flowering plants, the pollen grains and ovules are developed in specialized structures called flowers.
- In angiosperms, the seeds are enclosed in fruits.
- The angiosperms are an exceptionally large group of plants occurring in wide range of habitats.
- They range in size from the smallest Wolffia to tall trees of Eucalyptus (over 100 metres).
- They provide us with food, fodder, fuel, medicines and several other commercially important products.
- They are divided into two classes: the dicotyledons and the monocotyledons.

Flower – the reproductive structure



Pollen grains are produced in the anther (part of stamen) and ovules in the ovary (part of pistil).



Thus, in angiosperms, seeds are enclosed within fruits.

Angiosperms are divided into two classes

1. Dicotyledons (Dicots)

- Seeds have two cotyledons.
- Leaves usually with reticulate venation.
- Tap root system.
- Vascular bundles in stem arranged in a ring.
- Parts of flower usually in multiples of 4 or 5.

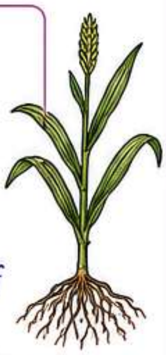
Reticulate venation

Floral parts in 4 or 5

Vascular bundles in a ring

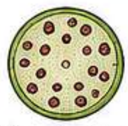
2. Monocotyledons (Monocots)

- Seeds have one cotyledon.
- Leaves usually with parallel venation.
- Fibrous root system.
- Vascular bundles in stem scattered.
- Parts of flower usually in multiples of 3.



Parallel venation

Floral parts in 3



Vascular bundles scattered

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